Renewable Variability and Electricity Price Risk in Spain

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Abstract

Wind and solar output changes over time, so other resources must adjust to meet electricity demand. I study how these changes relate to price risk using hourly electricity and weather data from Spain during 2015–2018. I use residual load to describe the demand remaining after wind and solar generation, construct a descriptive risk index, and compare models that predict high-price hours six hours ahead. In 2018, high-demand hours had a mean price of 70.41 EUR/MWh when wind–solar output was low and 66.04 EUR/MWh when it was high. Gradient boosting achieved the highest validation average precision (AP), but its test AP was 0.638, below the price-history baseline's 0.740. Full logistic regression achieved a test AP of 0.761. These findings show an association between residual load and price, while the value of additional predictors varies across years and models. The alarm analysis also shows that capturing more high-price events requires more frequent warnings and more false alarms.

1 Introduction

Electricity supports many everyday activities, but its price can change considerably within a day. I am interested in this variation because wind and solar output depends on conditions that people cannot fully control, while homes, schools, and businesses need electricity at particular times. This raises a question: when renewable output falls and demand stays high, which patterns suggest greater supply pressure and a higher chance of expensive hours?

Other resources, including conventional generation, hydropower, storage, and imports, must meet the demand remaining after wind and solar generation. Their availability and costs also influence prices. I therefore examine observed market data to understand how renewable output and prices vary together. Hourly observations make the timing of demand, generation, and prices easier to study than annual totals alone.

The Spanish dataset contains four years of national electricity measurements and city-level weather observations[1]. I focus on onshore wind and solar because the dataset records their generation separately for each hour. These series allow me to compare daily and seasonal patterns and calculate how much demand remains for other supply resources.

I examine three related questions. How do prices differ across combinations of demand and wind–solar output? Can generation, demand, and weather improve six-hour-ahead high-price warnings beyond recent prices and calendar information? How does the balance between false alarms and missed expensive hours change when missed events receive a higher cost?

I first compare observed market conditions and then test whether earlier observations can predict a high-price hour. The forecasting experiment follows time order, with later years reserved for validation and testing. I define high prices using a fixed threshold from the training period so that changes in event frequency remain visible. Finally, I compare alarm policies to examine the trade-off between frequent warnings and missed events, which may matter when scheduling flexible electricity use.

2 Data preparation

I use a public dataset with 35,064 hourly electricity records and 178,396 weather records from 2015–2018[1]. The source project attributes the electricity data to ENTSO-E and the weather observations to OpenWeather. The weather observations cover Madrid, Barcelona, Valencia, Seville, and Bilbao[2]. Kaggle lists the dataset as version 1 under a CC0 public-domain license.

The electricity variables are actual load, onshore wind generation, solar generation, and the field labeled `price actual`, which I call the reported actual price. Power is measured in MW and price in EUR/MWh. The weather variables are temperature, wind speed, humidity, cloud cover, and pressure. I use UTC timestamps to keep the hourly sequence continuous through daylight-saving changes. I use local Spanish time for calendar features and yearly data splits.

Data check	Count
Electricity observations	35,064
Weather observations	178,396
Repeated city-hour rows	3,076
Missing demand measurements	36
Missing onshore wind measurements	18
Missing solar measurements	18
Out-of-range wind-speed readings	2
Out-of-range pressure readings	58
Hours excluded from physical summaries	41

Table 1: Data quality checks. All reported actual prices are present. Temperature, humidity, and cloud cover have no out-of-range readings under the specified bounds, and every hour has observations from all five cities.

I average repeated weather readings within each city-hour and then give each of the five cities equal weight. I treat readings outside the following bounds as missing: 180–340 K for temperature, 0–60 m/s for wind speed, 0–100% for humidity and cloud cover, and 870–1085 hPa for pressure. I carry the last available observation forward for at most six hours and fill any remaining predictor gaps with training-set medians. The physical summaries exclude 41 hours with missing original load or wind–solar measurements.

Year	Hours	Load	Wind	Solar	Share	Price	High	Joint
2015	8,728	28.36	5.48	1.46	24.6	61.38	15.6	10.6
2016	8,779	28.51	5.42	1.41	24.2	47.44	4.4	3.5
2017	8,758	28.86	5.38	1.50	24.0	59.32	12.6	7.9
2018	8,758	29.06	5.58	1.37	24.0	63.44	18.1	10.2

Table 2: Annual summaries for observed physical measurements. Load, wind, and solar are mean GW; share is the mean wind–solar contribution to demand (%); price is mean EUR/MWh. High is the percentage above 73.53 EUR/MWh, and joint is the percentage with both a high price and residual load above 25.22 GW.

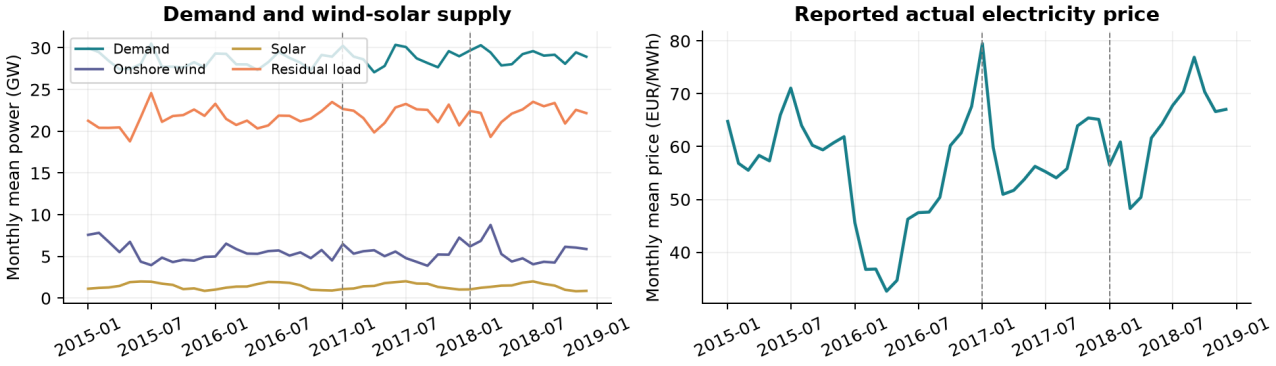


Figure 1: Monthly demand, generation, residual load, and reported actual price. Dashed lines mark the starts of the validation and test years.

3 Indicators and forecasting design

3.1 Pressure and economic-risk indicators

For hour t, let L_t, W_t, and S_t denote demand, onshore wind generation, and solar generation. Residual load is

$$N_t = L_t - W_t - S_t. \tag{1}$$

Residual load is the demand that other supply resources must meet. The dataset does not directly measure unmet demand, reserve adequacy, or outages, so I use the available observations to study supply pressure and price events.

I define a high-price hour as one with $P_t > q_P$, where $q_P = 73.53$ EUR/MWh is the training-period 90th percentile of reported actual price. I define high residual load using its training-period 75th percentile, $q_N = 25{,}215.5$ MW. A compound event occurs when both thresholds are exceeded. I also calculate the descriptive index

$$I_t = 0.45F_N(N_t) + 0.45F_P(P_t) + 0.10F_{\Delta P}(|P_t - P_{t-1}|) \tag{2}$$

Here, F_N, F_P, and $F_{\Delta P}$ are empirical distribution functions estimated from 2015–2016. The residual-load reference includes only observed physical measurements. Each function converts its input to a percentile rank, allowing supply pressure, price level, and absolute hourly price change to share a common scale. The weights specify how these three components contribute to the index. This index describes current conditions; the forecasting model estimates the probability of a future high-price hour.

3.2 Prediction target and chronological partitions

At origin t, the classifier predicts

$$Y_t = \mathbf{1}\{P_{t+6} > q_P\}, \qquad \widehat{p}_t = \Pr(Y_t = 1 \mid X_t). \tag{3}$$

I use 31 predictors. The price features include the current price, prices 1, 24, and 168 hours earlier, the trailing 24-hour mean and standard deviation, and the hourly price change. Demand, wind, solar, and residual load each contribute their current value, one-hour change, and 24-hour lag. I also include the wind–solar share, five weather averages, and sine–cosine representations of the target hour, weekday, and month. All measured predictors use observations available through t. The experiment assumes that these observations are available when the forecast is made; publication delays are not specified in the source.

Partition	Origin years	Forecasts	Events	Event rate (%)
Train	2015–2016	17,370	1,701	9.79
Validation	2017	8,754	1,108	12.66
Test	2018	8,754	1,587	18.13

Table 3: Chronological forecasting partitions after the 168-hour lag burn-in and removal of targets crossing a year boundary. The price-event threshold remains fixed across partitions.

3.3 Models and evaluation

I compare five models: calendar-only logistic regression, logistic regression using price history and calendar information, full logistic regression, a random forest, and histogram-based gradient boosting[3]. The logistic models use training-set medians for missing predictors and standardize the inputs. The forest has 150 trees, a maximum depth of 12, and at least 20 samples per leaf. Boosting uses 160 iterations, a learning rate of 0.06, at most 15 leaves, at least 50 samples per leaf, and ℓ_2 regularization of 1. To examine the contribution of different features, I also fit boosting models without weather or without wind–solar and residual-load features.

I keep the model settings fixed and select the forecasting model with the highest 2017 average precision among the price-history baseline and the three full models. For each model, I choose an alarm threshold between 0.01 and 0.99 that maximizes validation F_1. The models remain fitted on 2015–2016. I assess test performance using average precision (AP), ROC–AUC, Brier probability error, and alarm precision, recall, and F_1. I use paired seven-day bootstrap blocks to estimate uncertainty in the selected model's comparison with the price-history baseline.

4 Market patterns and renewable variability

Table 2 and Figure 1 show that mean demand rises from 28.36 GW in 2015 to 29.06 GW in 2018, while the mean wind–solar share stays close to 24%. Prices vary more strongly: the annual mean falls to 47.44 EUR/MWh in 2016 and rises to 63.44 EUR/MWh in 2018. These differences suggest that the annual renewable share alone does not fully describe market conditions.

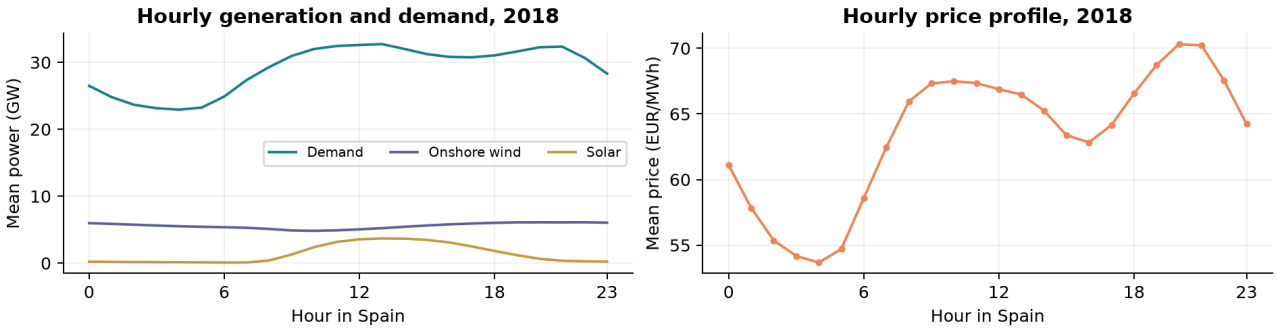


Figure 2: Average hourly demand, onshore wind, solar generation, and price in 2018, expressed in local Spanish time.

Figure 2 shows how demand, generation, and prices vary over a typical day in 2018. Solar output is concentrated around midday, while demand remains high into the evening. Average prices are lowest near 04:00 and highest near 20:00. The evening therefore stands out as a period when falling solar output coincides with relatively high demand and prices. These averages summarize many different weather and market conditions.

I divide the 2018 observations into four groups using the training medians of demand, 28.69 GW, and wind–solar output, 6.46 GW. A value above its median belongs to the high group. During high-demand hours, low wind–solar output is associated with a mean residual load of 27.81 GW and a mean price of 70.41 EUR/MWh. When output is high, the corresponding values are 23.39 GW and 66.04 EUR/MWh. High-price frequency is also lower, falling from 33.0% to 23.1%.

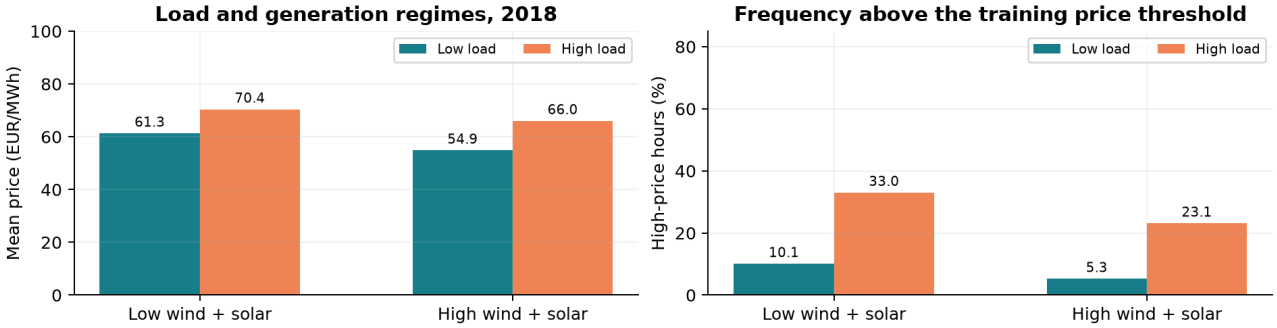


Figure 3: Mean price and high-price frequency across demand and wind–solar regimes in 2018. The thresholds defining each regime come from the training years.

The low-demand groups show a similar pattern. Mean price is 61.31 EUR/MWh with low wind–solar output and 54.86 EUR/MWh with high output. High-price frequency is 10.1% and 5.3%, respectively. These comparisons suggest that renewable availability, residual load, and price are related. They do not establish a causal effect because the groups also differ in timing and other supply conditions.

Across hours with observed physical measurements, the Spearman correlation between residual load and price is 0.506. The correlations with price are 0.436 for demand, −0.220 for onshore wind, and 0.133 for solar. The positive solar correlation needs to be read alongside the daily pattern: solar generation occurs during hours that also have substantial demand. Together, the hourly profiles and group comparisons provide context for interpreting these correlations.

5 Forecasting results

Model	Val. AP	Test AP	AUC	Brier	Prec.	Recall	F_1	τ
Calendar	0.288	0.368	0.749	0.142	0.348	0.577	0.434	0.12
Price + calendar	0.842	0.740	0.909	0.087	0.699	0.616	0.655	0.35
Full logistic	0.849	0.761	0.917	0.084	0.677	0.683	0.680	0.29
Full forest	0.862	0.705	0.905	0.086	0.537	0.807	0.645	0.24
Full boosting	0.879	0.638	0.895	0.098	0.618	0.641	0.630	0.28
No weather	0.875	0.629	0.892	0.100	0.626	0.611	0.618	0.30
No wind–solar	0.879	0.612	0.879	0.104	0.558	0.658	0.604	0.22

Table 4: Validation AP and test metrics for six-hour-ahead high-price forecasts. All remaining columns describe the 2018 test set. AUC denotes ROC–AUC, and τ is the alarm threshold selected on validation data. The two removal models use the full boosting settings. Lower Brier scores indicate smaller probability errors.

The full boosting model achieves the highest validation AP, 0.879, so I select it before examining the test results. Its test AP falls to 0.638, below the price-history baseline's 0.740. Full logistic regression achieves a test AP of 0.761 and the lowest test Brier score, 0.084. The model that performs best in validation therefore does not perform best in the following year.

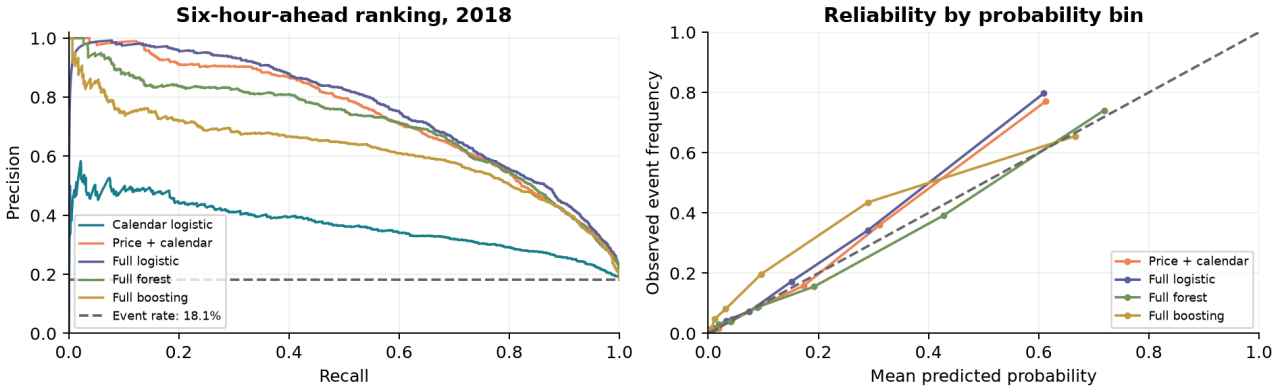


Figure 4: Test precision–recall curves and reliability curves. The dashed horizontal line in the left panel is the test event rate; the diagonal in the right panel represents agreement between predicted probabilities and observed event frequencies. Reliability bins contain approximately equal numbers of forecasts.

Figure 4 shows that models using recent prices rank high-price events more effectively than the calendar-only model. Several points on the reliability curves lie above the diagonal, meaning that the observed event frequency exceeds the predicted probability in those bins. The proportion of high-price targets also changes over time, rising from 9.8% in training to 12.7% in validation and 18.1% in testing. This shift provides context for the weaker test performance.

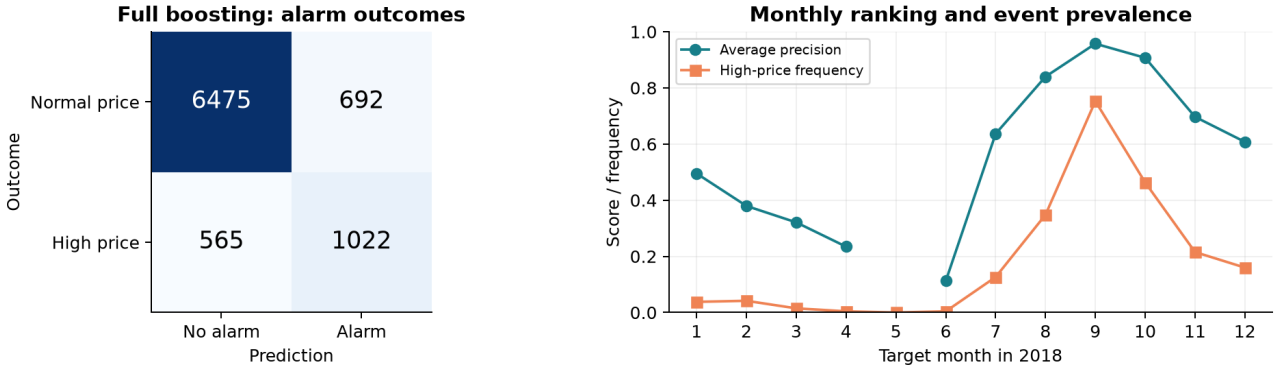


Figure 5: Alarm outcomes for the validation-selected boosting model at $\tau = 0.28$ and its monthly test AP. May has no high-price events, so its monthly AP is undefined and appears as a gap.

The selected model produces 1,646 alarms for 8,754 forecasts. It correctly flags 1,018 events, produces 628 false alarms, and misses 569 events, giving 61.8% precision and 64.1% recall. High-price events concentrate in late summer and autumn (Figure 5). September has a 75.3% event frequency and AP of 0.958, while several spring months have few events. Monthly event frequency matters when interpreting AP.

6 Information use and alarm decisions

Quantity	Estimate	2.5th percentile	97.5th percentile
Selected model AP	0.638	0.537	0.725
Price + calendar AP	0.740	0.621	0.800
Paired AP difference	-0.102	-0.182	-0.006

Table 5: Paired seven-day bootstrap comparison, using 250 resamples and fitted models held fixed.

The selected model's test AP is 0.102 below the price-history baseline's. The bootstrap interval is $[-0.182, -0.006]$ (Table 5). Removing weather or wind–solar features gives test AP values of 0.629 and 0.612. Both groups offer modest benefits within boosting.

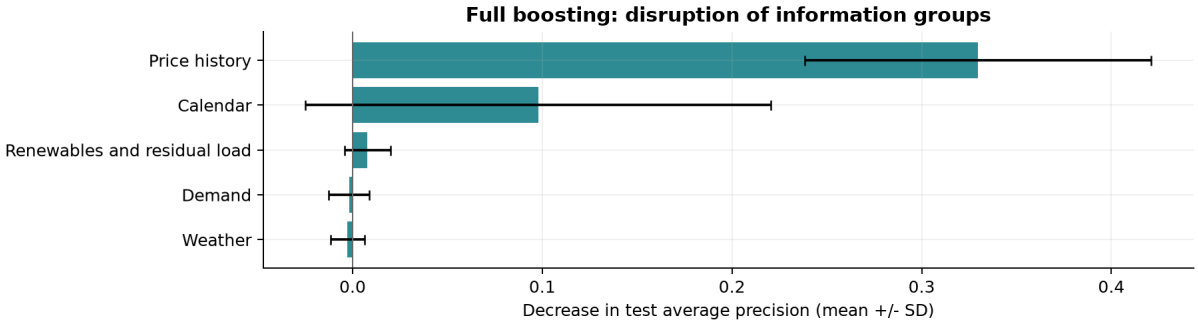


Figure 6: Full boosting AP changes after whole-day circular shifts. Bars show ten-shift means and one standard deviation.

Figure 6 shows that shifting price history reduces AP by 0.351 on average, making it the main information source. Calendar shifts reduce AP by 0.107, with substantial variation. Wind–solar and weather effects are small, and the demand effect is near zero. These comparisons describe model dependence without establishing causal effects.

I give false alarms a relative cost of 1 and missed events a cost of r. I choose each threshold by minimizing validation loss $(\mathrm{FP} + r\mathrm{FN})/n$. These hypothetical costs express preferences about warning errors and do not estimate financial savings.

r	τ	Alarms (%)	Recall (%)	FP/1,000	FN/1,000	Loss
1	0.31	17.5	60.5	65.0	71.6	0.137
3	0.17	24.8	75.5	110.9	44.4	0.244
5	0.10	30.2	82.0	153.9	32.7	0.317
10	0.06	35.9	88.3	199.5	21.2	0.412
20	0.05	37.9	89.4	217.5	19.3	0.604

Table 6: Test outcomes at validation-selected thresholds. FP and FN denote false alarms and missed events; loss uses each row's ratio.

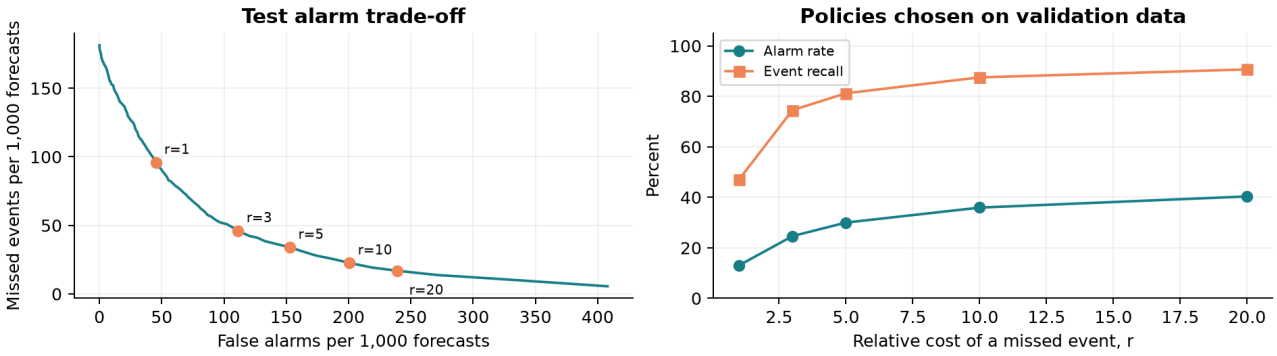


Figure 7: Test alarm trade-offs. Markers identify validation-selected cost-ratio policies from Table 6.

Increasing r from 1 to 10 raises recall from 60.5% to 88.3% and the alarm rate from 17.5% to 35.9%. Missed events fall from 71.6 to 21.2 per 1,000 forecasts, while false alarms rise from 65.0 to 199.5.

7 A high-price week and concluding discussion

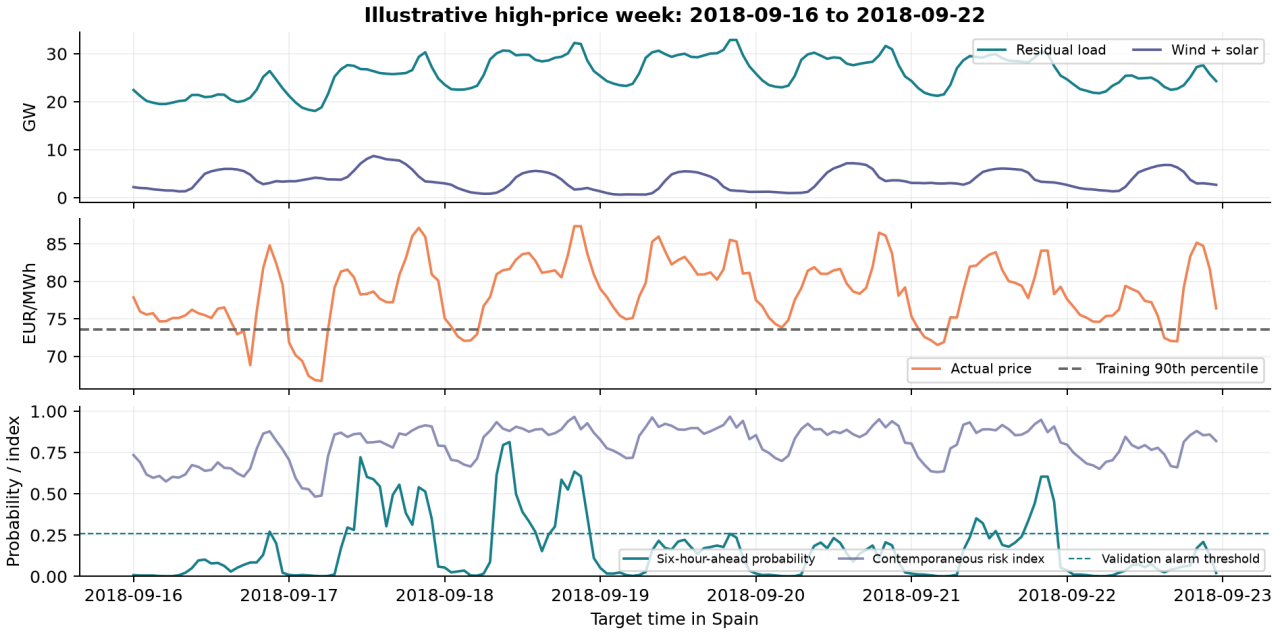


Figure 8: The earliest seven-day window with the largest number of high-price hours in 2018. Forecast probabilities are plotted at their target times and were issued six hours earlier. The risk index uses contemporaneous observations.

The week from 16 to 22 September contains 147 high-price hours out of 168. Figure 8 shows high prices alongside changing residual load and wind–solar output. The descriptive risk index stays high for much of the week. Forecast probabilities, however, sometimes fall below the alarm threshold even though the later price exceeds 73.53 EUR/MWh. This example shows how the model can miss events during a period of persistently high prices. I select the week because it has a high concentration of events; the full-year results provide the broader assessment.

The market comparisons show a consistent association: high-demand hours with low wind–solar output have higher mean residual load, higher mean prices, and more frequent high-price events than high-demand hours with greater renewable output. The daily profiles also highlight the evening, when solar generation declines while demand and prices remain relatively high. These findings describe the changing demand that other supply resources must meet.

The forecasting results are less uniform. Full logistic regression achieves a slightly higher test AP than the price-history baseline, but the boosting model selected on validation data performs worse in testing. Price history provides most of the boosting model's ranking information. The benefit of adding demand, generation, and weather therefore depends on the model, and changes in annual price conditions matter for interpreting warning performance.

The alarm analysis shows the practical trade-off between catching more high-price hours and issuing more false alarms. Forecast probabilities and alarm counts can help users consider how often they would accept a warning when scheduling electricity use. The available data support an analysis of residual load and price risk. Assessing operational adequacy or actual cost reductions would require additional information.

References

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